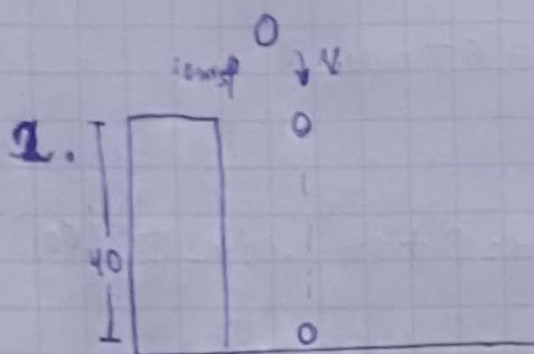


Parcial Física



$$V_i = 10 \text{ m/s}$$

$$h_{\text{max}} = \frac{10^2}{2(9,8)}$$

$$= 5,1 \text{ metros}$$

$$t_{\text{max}} = \frac{10}{9,8} = 1,02 \text{ seg}$$

$$h_{\text{total}} = 40 + 5,1 = 45,1 \text{ m}$$

$$h = v_i t + \frac{1}{2} g t^2$$

$$45,1 = \cancel{0 \cdot t} + \frac{1}{2} (9,8) \cdot t^2$$

$$45,1 = 4,9 \cdot t^2$$

$$9,2 = t^2$$

$$t = 3,03 \text{ seg}$$

$$\therefore t_{\text{total}} = 3,03 + 1,02 = 4,05 \text{ seg}$$

total.

4.

$$180 \text{ RPM} = \frac{180 \text{ Vueltas}}{\text{m}} \cdot \frac{\text{m}}{60} \cdot \frac{2\pi \text{ rad}}{\text{Vueltas}} = 6\pi \text{ rad/s}$$

$$\omega_f = \omega_i - \alpha \cdot t$$

$$\Rightarrow 0 = 6\pi \text{ rad} - \frac{\pi}{10} \text{ rad} \cdot t$$

$$t = \frac{6\pi}{\frac{\pi}{10}}$$

$$t = 60 \text{ seg}$$

$$\theta = 0 + 6\pi \text{ rad} \cdot 60 + \frac{1}{2} \left(\frac{\pi}{10} \right) (60)^2$$

$$\theta = 540 \pi \text{ rad} \cdot \frac{\text{Vueltas}}{2\pi \text{ rad}}$$

$$\theta = 270 \text{ vueltas}$$